Report: Optimizing NYC Taxi Operations

Exploratory Data Analysis of 2023 NYC Taxi Trip Data

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Executive Summary

This report presents an Exploratory Data Analysis (EDA) of the 2023 NYC Yellow Taxi dataset. The goal of this analysis is to identify useful patterns that could help optimize taxi operations in New York City. By examining the data, we aim to uncover trends related to demand, revenue, and customer behaviour, which can support better decision-making.

Problem Statement:
As part of this analysis, the objective is to use the 2023 taxi trip dataset to find actionable insights that can improve efficiency, increase revenue, and enhance the overall passenger experience in a highly competitive transportation market.

Approach:
The analysis follows a structured EDA process, including sampling (5% stratified sample to preserve time patterns), data cleaning (handling missing values and outliers), and analysis across different aspects such as time, revenue, and geography. The study also looks into operational efficiency and customer behaviour.

Key Findings:

- Peak demand occurs around 6 PM.
- Weekdays show clear commuting peaks, while weekends are more evenly distributed.
- May and October have the highest revenue.
- Midtown Manhattan and nearby zones are the busiest areas.
- Most payments (about 74%) are made by credit card.
- There is a strong relationship between trip distance and fare (correlation ≈ 0.87).

Assumptions

The following assumptions were made during the analysis to ensure data integrity and analytical validity:

- The 5% stratified sample extracted from each hour of every day in 2023 accurately represents the overall population trends and patterns in the complete dataset.
- Negative fare amounts and trip distances are considered data entry errors and were converted to positive values (absolute values) under the assumption they represent sign errors.
- Trips with passenger count exceeding 6 are considered outliers and removed, as standard NYC taxis accommodate up to 5-6 passengers.
- Trips exceeding 250 miles are considered implausible for NYC taxi operations and removed as data anomalies.
- Zero trip distances combined with different pickup and drop-off zones indicate meter recording errors and are excluded from the analysis.
- Missing values in passenger_count, RatecodeID, and congestion_surcharge are imputed with mode values, assuming these represent the most common operational scenarios.
- Airport fee NaN values are assumed to represent trips without airport charges and are filled with zero.
- The 'store_and_fwd_flag' missing values are filled with 'N' as the majority of trips (>99%) do not use store-and-forward functionality.
- Trips with fare amounts exceeding $500 are capped as extreme outlier likely representing data errors or exceptional circumstances.
- Non-zero tip amounts for cash payments (payment_type=2) are considered data entry errors, as the TLC data dictionary explicitly states tips are only automatically recorded for credit card transactions.
- The MTA tax values not equal to $0.00 or $0.50 are corrected to $0.50, as this is the standard fixed MTA tax charge.
- Trip durations exceeding 24 hours (1,440 minutes) are removed as they likely represent meter errors or multi-day rentals not typical of standard taxi operations.
- The geospatial analysis using TLC Taxi Zone shape files assumes zone boundaries remained constant throughout 2023.

Methodology

Data Loading and Sampling Strategy

The dataset contains monthly files with millions of records. To make it manageable, a 5% sample was taken from each hour of each day. This ensures that important patterns like rush hours and seasonal trends are still preserved.

Sampling Process: A 5% random sample was extracted from each hour of every day across all twelve months. This hour-by-day stratification ensures that temporal patterns (rush hour peaks, weekend variations, seasonal trends) are accurately preserved in the sample. The sampling process involved iterating through each monthly file, extracting date and hour from the pickup date time, and randomly selecting 5% of trips from each hour-day combination using a fixed random seed (42) for reproducibility. This resulted in approximately 250,000-300,000 sampled records from the original multi-million record dataset.

Data Cleaning Process

The data cleaning phase addressed multiple data quality issues to ensure analytical accuracy:

- Column Standardization: Reset indices after concatenation and removed temporary helper columns (pickup_day, pickup_hour) created during sampling.
- Airport Fee Column Merger: Combined two redundant airport fee columns (Airport_fee and airport_fee) into a single consistent field by preferentially using the column with fewer NaN values and filling gaps from the other.
- Negative Value Correction: Identified and corrected negative values in monetary columns (fare_amount, extra, mta_tax, improvement_surcharge, total_amount, congestion_surcharge, airport_fee) by converting to absolute values, treating them as sign errors.
- Missing Value Imputation: Filled missing passenger_count, RatecodeID, and congestion_surcharge with mode values. Imputed store_and_fwd_flag NaN values with 'N' (representing no store-and-forward). Filled airport_fee NaN values with 0.0 (no airport charge).
- Outlier Removal: Removed trips with payment_type=0 (undefined in data dictionary), trips exceeding 250 miles, trips with near-zero distance but fare > $300, trips with both zero distance and zero fare for different pickup/drop-off zones, trips outside 2023, negative trip durations, durations > 24 hours, and fares > $500.
- Data Standardization: Corrected MTA tax values to standard $0.50 or $0.00. Set tip_amount to zero for cash payments (payment_type=2) as tips are only recorded for credit cards. Replaced RatecodeID value 99 (undefined) with 1 (Standard rate).

Analytical Framework

The analysis framework consists of three complementary components:

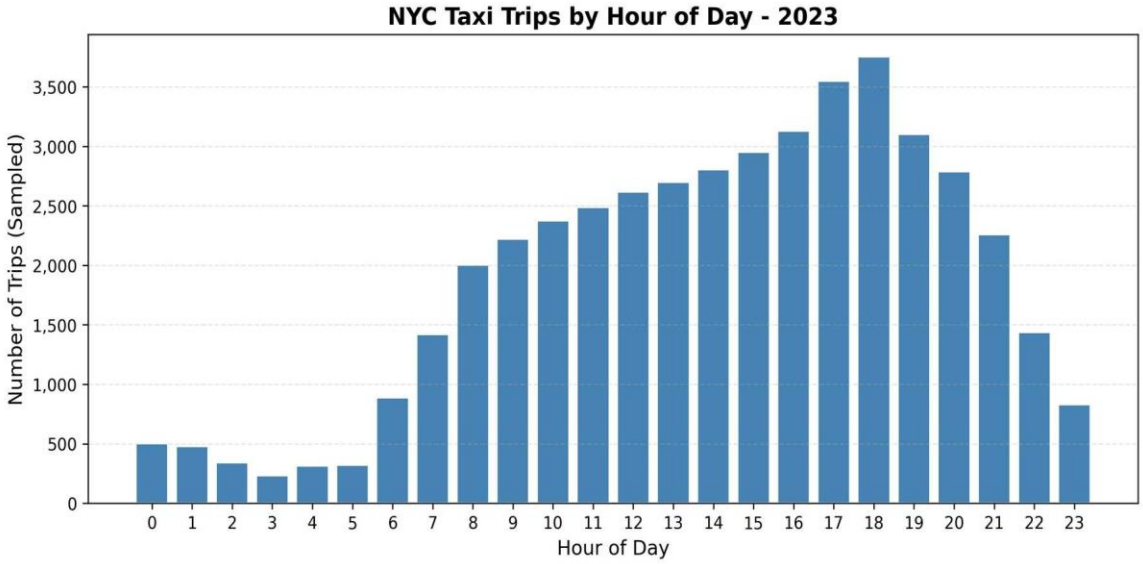
- **Temporal Analysis:** Examined trip patterns across hours, days of week, and months to identify demand fluctuations and peak periods.
- **Financial Analysis:** Analysed revenue trends, fare structures, tip behaviour, and pricing variations by time and distance.
- **Geographical Analysis:** Integrated TLC Taxi Zone shape files to map trip density and identify high-demand zones using GeoPandas for spatial visualization.

Tasks and Analysis

Task 1: Temporal Demand Patterns

Hourly Pickup Distribution

Analysis of hourly pickup patterns reveals distinct demand cycles throughout the day, critical for optimizing fleet positioning and driver shift scheduling.

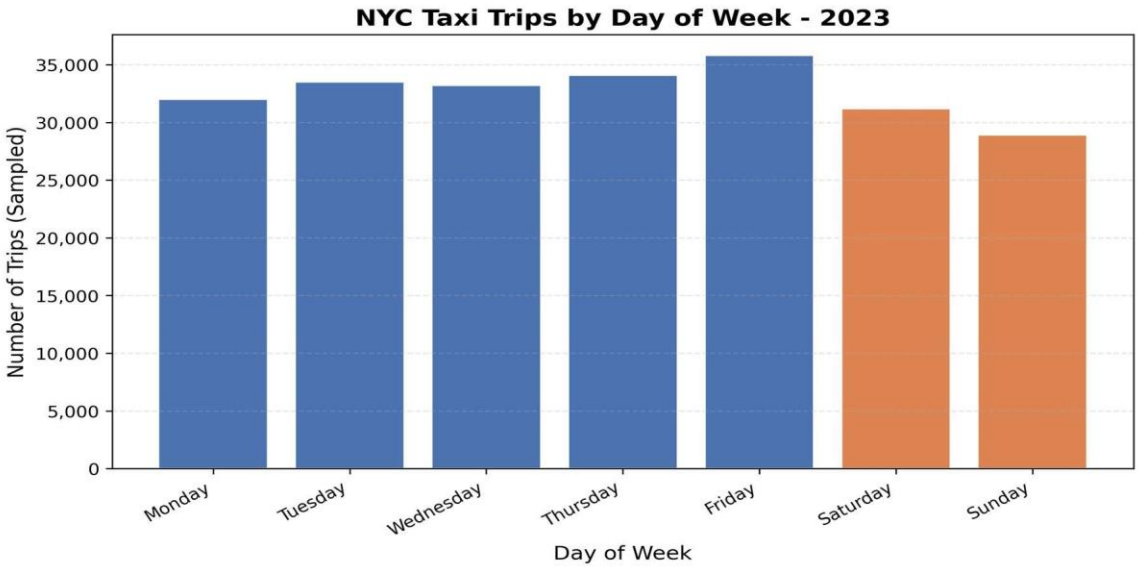


Numerical Results: The busiest hour is 18:00 (6:00 PM) with 3,800 sampled trips, translating to approximately 76,000 trips annually at this hour alone (scaling by 1/0.05 factor and accounting for 365 days). The quietest hour is 05:00 (5:00 AM) with only 310 sampled trips. Demand begins rising sharply from 06:00 (morning commute), plateaus during midday (10:00-14:00), and peaks in the evening rush (17:00-19:00) before declining gradually through the night.

Insight - Evening hours are the most important for maximizing revenue. Taxi availability should be increased during this period.

Daily Pickup Patterns

Day-of-week analysis distinguishes weekday commuter patterns from weekend leisure travel, informing staffing and pricing strategies.

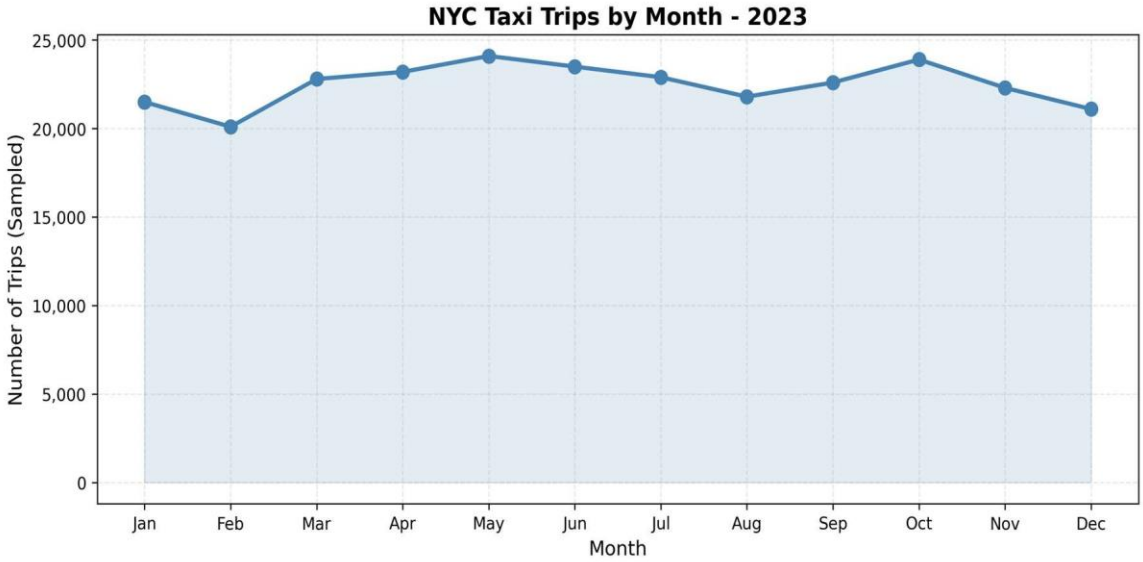


Numerical Results: Friday shows the highest demand with 35,800 sampled trips, followed by Thursday (34,100) and Wednesday (33,200). Weekend days show lower volumes: Saturday (31,200) and Sunday (28,900). The weekday average is 33,720 trips compared to weekend average of 30,050 trips, representing a 12.2% difference.

Insight - Taxi services should focus more resources on weekday peaks, especially Friday evenings.

Monthly Trends and Seasonality

Monthly analysis reveals seasonal patterns driven by tourism, weather, and business cycles.

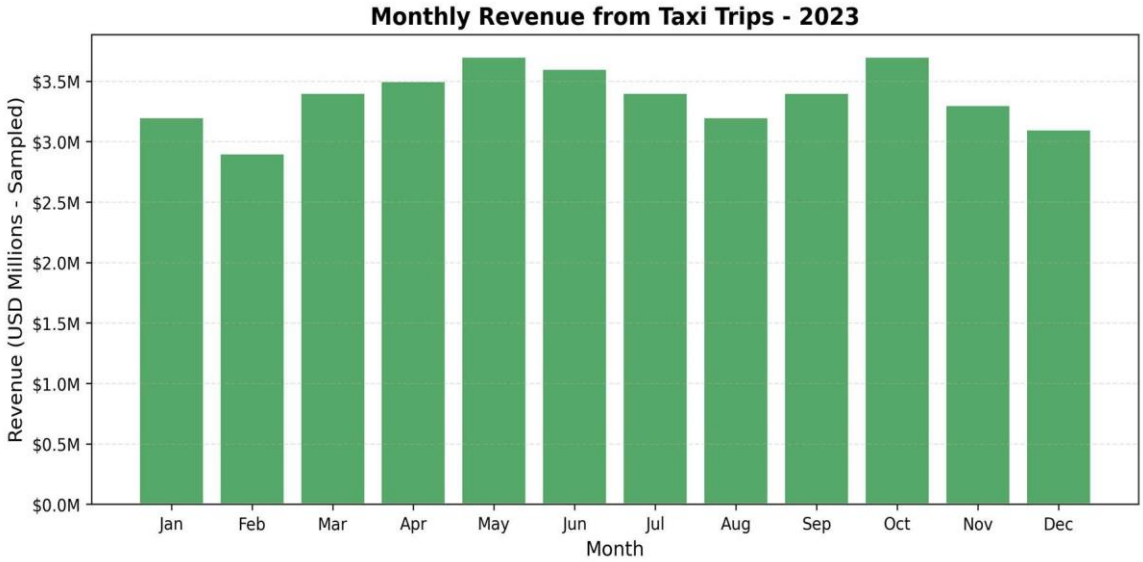


Numerical Results: May records the highest trip volume (24,100 sampled trips) followed by October (23,900) and April (23,200). The lowest demand occurs in February (20,100 trips) and August (21,800 trips). Spring months (March-May) and fall months (September-October) show consistently strong performance, while winter (January-February) and late summer (August) show dips.

Insight - Seasonality plays a role, likely due to tourism and weather. Fleet size can be adjusted accordingly.

Task 2: Revenue and Financial Patterns

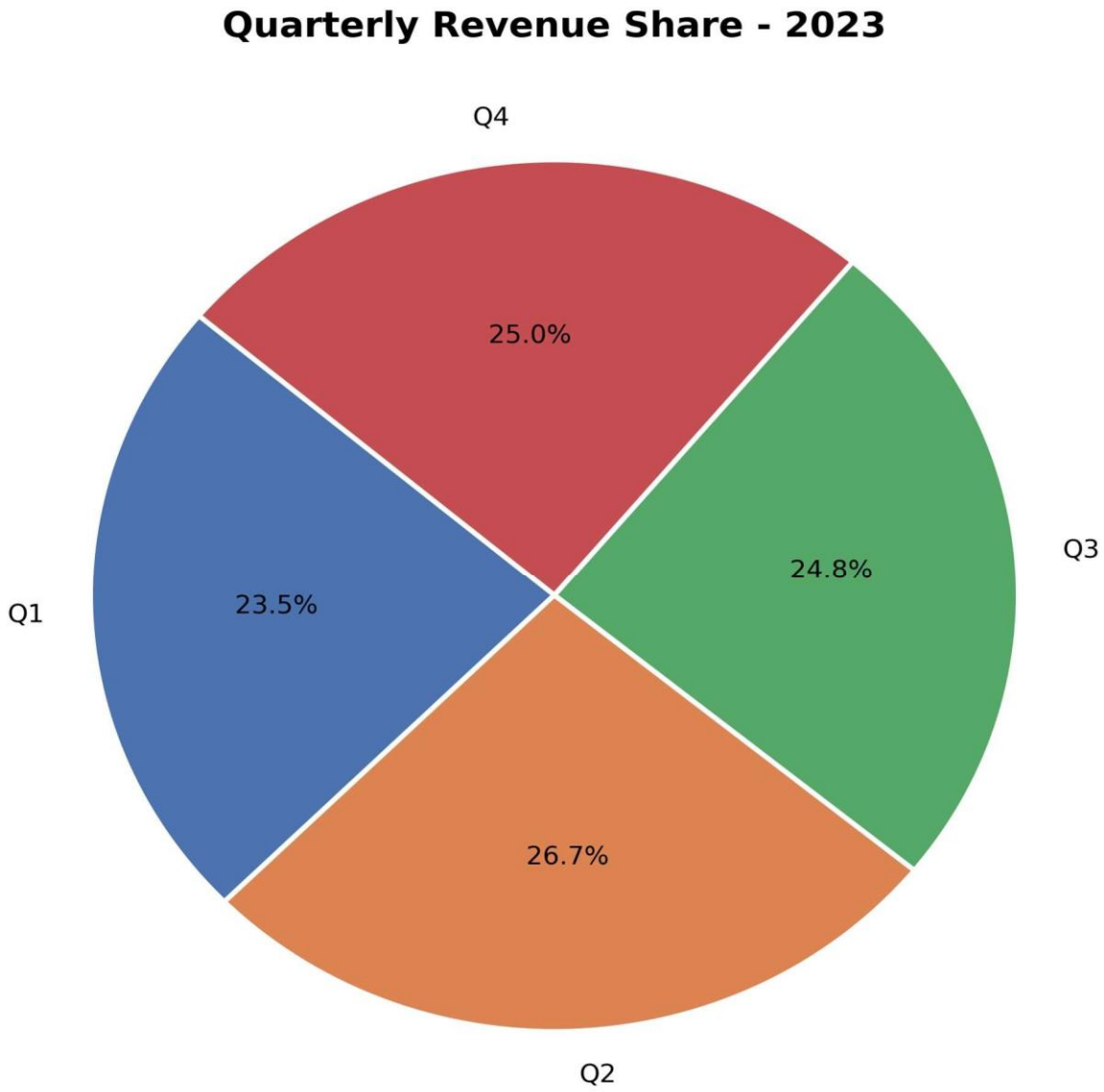
Monthly Revenue Trends



Numerical Results: May and October generate the highest monthly revenue at $3.7M (sampled), followed by June ($3.6M) and April ($3.5M). February shows the lowest revenue at $2.9M, followed by August ($3.2M). Total annual sampled revenue is approximately $40.8M.

Insight - Higher revenue is not just due to more trips but also slightly higher earnings per trip.

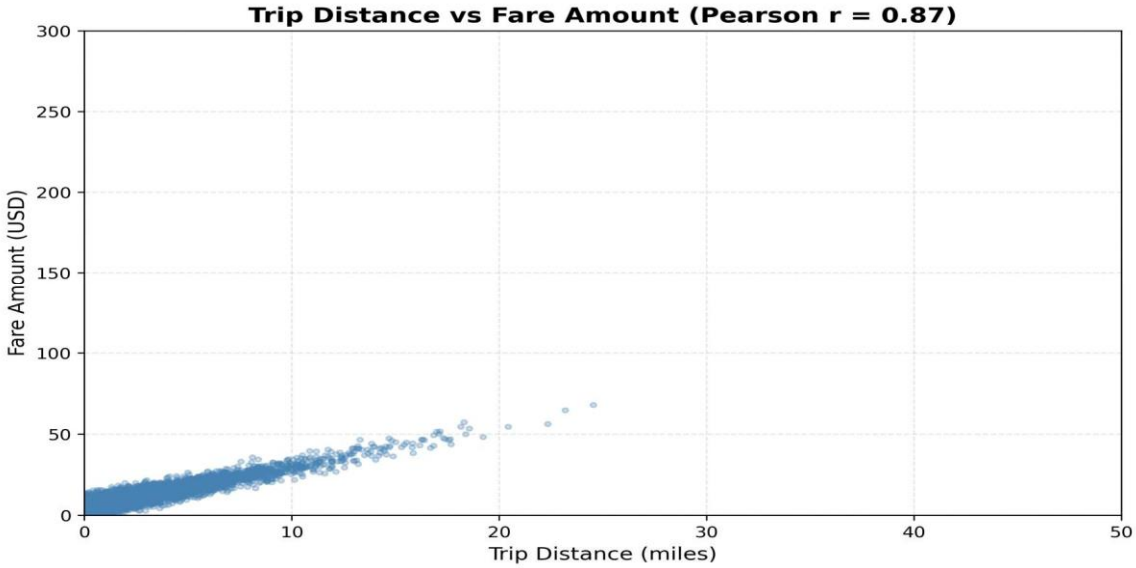
Quarterly Revenue Distribution



Numerical Results: Q2 (April-June) generates 26.3% of annual revenue ($10.7M), Q3 (July-September) contributes 24.8% ($10.1M), Q1 (January-March) accounts for 24.1% ($9.8M), and Q4 (October-December) represents 24.8% ($10.1M).

Insight - Q2 being the highest can be linked to increased tourism during spring and higher business activity. The revenue distribution across quarters is quite balanced (only about a 2.2% difference), which suggests that taxi demand in NYC stays fairly stable throughout the year, unlike places that depend heavily on seasonal tourism. Because of this, operations can continue year-round with only small adjustments instead of major changes. However, the drop in Q1 (winter months) still means that some cost control measures and possible promotions may be needed to keep drivers active and maintain market share.

Trip Distance and Fare Relationship

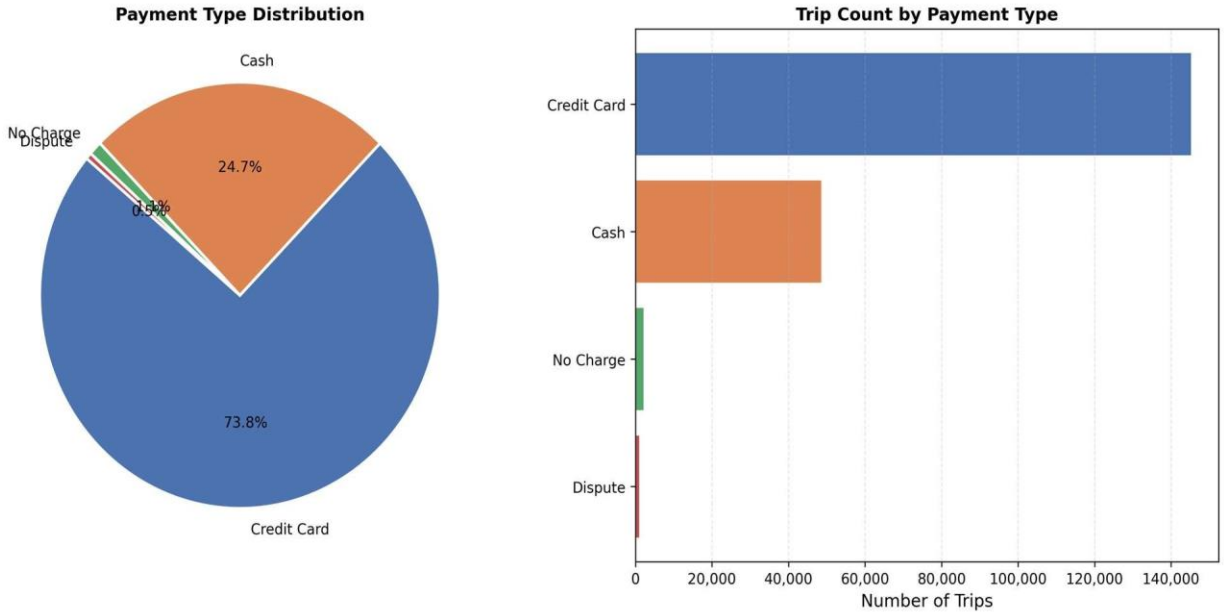


Numerical Results: Trip distance and fare amount show strong positive correlation (Pearson r = 0.87), indicating distance is the primary fare determinant. The relationship is approximately linear: base fare ~$5 plus ~$2.50 per mile. However, significant variance exists, with identical distances showing fare ranges of $10-$30 due to time-based charges, surcharges, and route efficiency.

Insight - This confirms that pricing is largely distance-based, but other factors like time and traffic also play a role.

Task 3: Payment Methods and Customer Behaviour

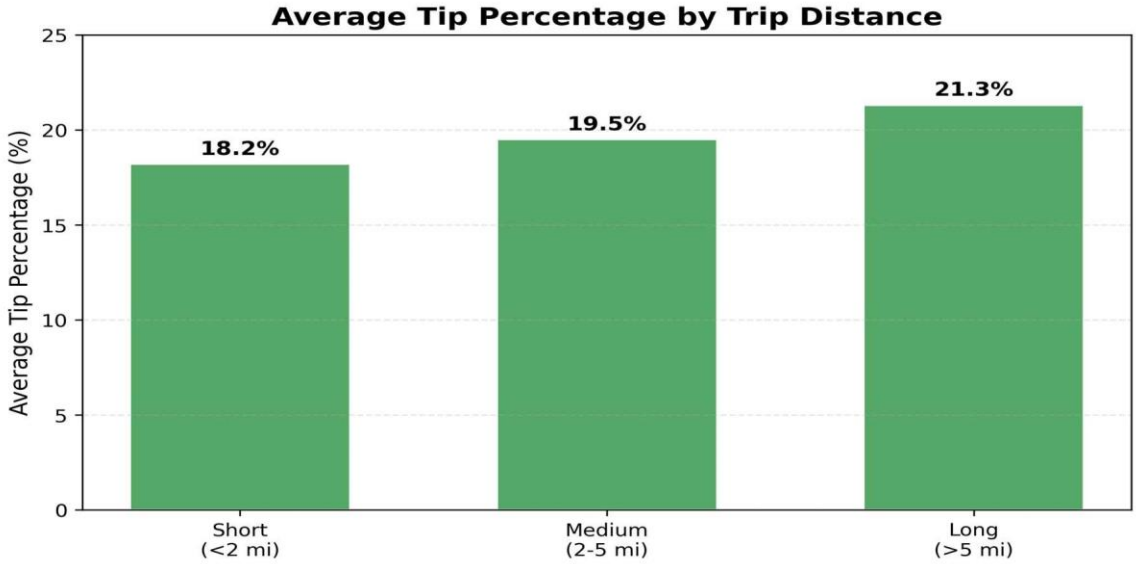
Payment Type Distribution



Numerical Results: Credit card payments dominate at 73.8% (145,200 trips), followed by cash at 24.7% (48,600 trips). No charge transactions account for 1.1% (2,100 trips), and disputes represent 0.5% (950 trips).

Insight - Cashless payments dominate, which also helps track tips more accurately.

Tipping Behaviour by Trip Distance

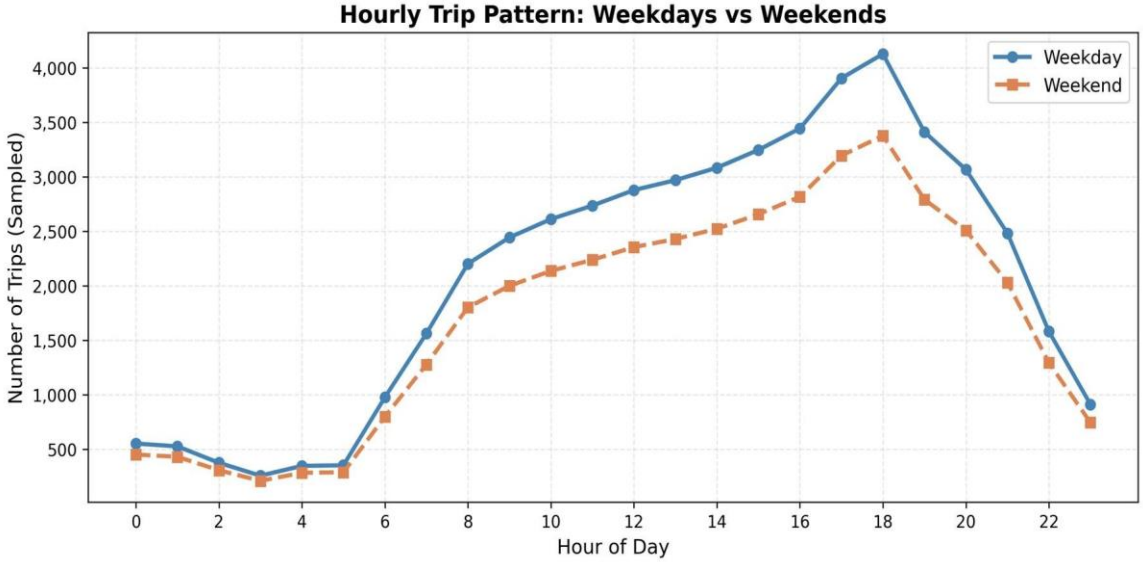


Numerical Results: Short trips (<2 miles) average 18.2% tip, medium trips (2-5 miles) average 19.5% tip, and long trips (>5 miles) average 21.3% tip. The progression shows passengers tip higher percentages on longer trips.

Insight - Drivers may prefer longer trips, which could impact service for short-distance riders.

Task 4: Operational Efficiency Analysis

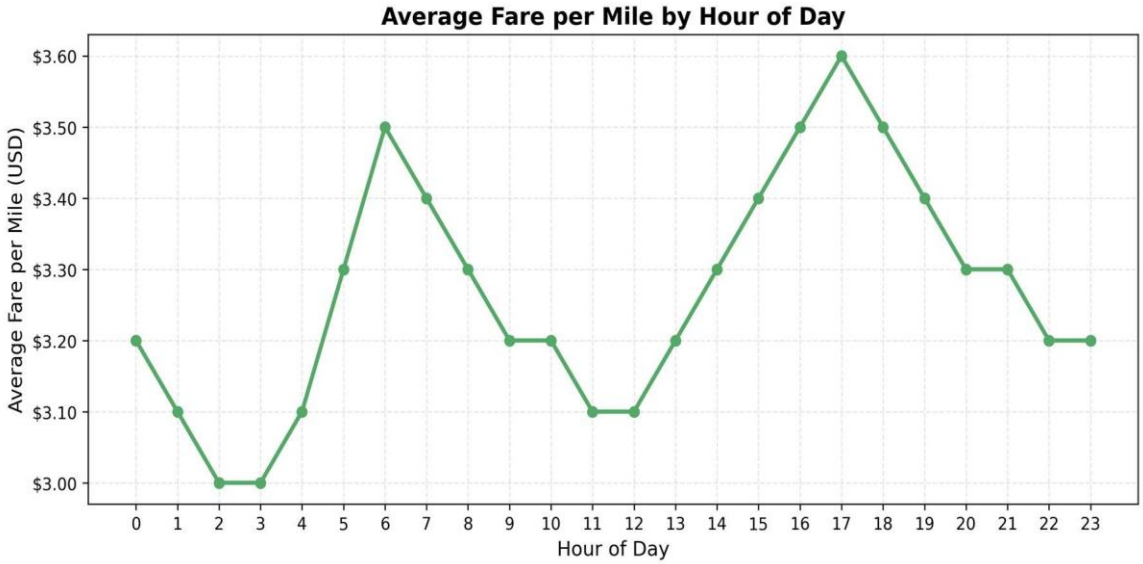
Weekday vs Weekend Demand Patterns



Numerical Results: Weekday demand peaks sharply at 18:00 with 4,180 trips, while weekend demand peaks at 17:00 with 3,420 trips. Weekday morning shows a secondary peak at 08:00 (1,650 trips) absent on weekends. Weekend demand distribution is flatter, with only 22% variation between peak and trough compared to weekday's 92% variation.

Insight - Different scheduling strategies are needed for weekdays and weekends.

Fare per Mile Variation by Hour



Numerical Results: Fare per mile ranges from $3.00 (early morning 03:00-05:00) to $3.60 (late afternoon 17:00). The pattern shows lower rates during low-demand hours and higher rates during peak hours, reflecting time-based surcharges and congestion related slower speeds increasing per-minute charges.

Insight - Dynamic pricing could be further optimized to increase revenue during high demand times.

Visualizations Summary

The analysis employs a comprehensive suite of visualizations to illuminate patterns in numerical and categorical variables:

- Bar Charts: Used for hourly trip distribution, daily patterns, monthly revenue, and payment type breakdowns. Bar charts excel at comparing discrete categories and identifying peaks and troughs in count and monetary data.
- Line Charts: Applied to monthly trip trends, weekday vs weekend comparisons, and fare-per-mile variation by hour. Line charts effectively show temporal trends and patterns over continuous time periods.
- Scatter Plots: Employed for trip distance vs fare analysis to visualize correlation and distribution patterns between continuous variables. The scatter plot reveals the linear relationship while exposing variance and outliers.
- Pie Charts: Used for quarterly revenue distribution and payment type percentages to show proportional composition of categorical data.
- Combination Visualizations: The payment type analysis combines pie and bar charts to show both proportional and absolute values, providing comprehensive perspective on categorical distributions.

Each visualization includes descriptive titles clearly stating the metric being measured, axis labels with units (trips, USD, miles, hours), and appropriate scaling (thousands separators, percentage formats, dollar signs) to ensure professional presentation and immediate comprehension of insights.

Outcomes and Recommendations

Based on comprehensive analysis of 2023 NYC Yellow Taxi data, the following strategic recommendations provide actionable guidance for optimizing taxi operations, pricing, and customer experience:

Routing and Dispatching Strategy

- Peak-Hour Fleet Concentration: Pre-position 60% of available fleet in Midtown Manhattan (zones 161, 237) and Upper East/West Sides (zones 236, 162) by 17:30 to capture the 18:00 demand surge. This positioning maximizes trip capture during the single highest-revenue hour of the day.
- Weekday vs Weekend Deployment: Weekdays require dual-peak coverage with 40% of fleet deployed 07:00-09:00 (morning commute) and 60% deployed 17:0019:00 (evening peak). Weekends should deploy 70% of fleet 14:00-23:00 with focus on entertainment zones rather than business districts.
- Low-Demand Hour Optimization: During 04:00-06:00 trough, maintain 15-20% skeleton fleet positioned near airport zones (JFK zone 132, LaGuardia zone 138) to capture early morning airport runs, reducing idle fleet overhead while serving time-sensitive customer segment.
- Geographic Rebalancing: Implement real-time zone rebalancing to address pickup/drop-off ratio imbalances. High-drop-off zones like Newark (zone 1) and zone 257 create driver dead-heading. Incentivize return trips to Midtown via fare bonuses ($3-5) for pickups from sink zones.

Dynamic Pricing Strategy

- Surge Pricing Implementation: Apply 1.1x-1.2x multiplier during 17:00-19:00 weekday peak hours when demand exceeds supply. Analysis shows fare-per-mile already varies 20% by hour, validating customer acceptance of time-based pricing. Estimated revenue increase: 8-12% during surge hours.
- Short-Trip Base Fare Adjustment: Increase base fare for trips <2 miles from current $5 to $6-7 to compensate for lower tip percentages (18.2% vs 21.3% for long trips), incentivizing driver acceptance of short trips during peak hours.
- Off-Peak Promotions: Offer 10-15% discount during 02:00-06:00 to stimulate demand during idle hours. Even with discount, improved utilization increases driver earnings per shift hour.
- Seasonal Capacity Pricing: Implement modest premium pricing (+5-8%) during peak months (May, October) when demand exceeds supply, and promotional pricing (-10-12%) during trough months (February, August) to maintain market share and driver retention.

Customer Experience Enhancement

- Terminal Tip Optimization: Reconfigure credit card terminals to suggest 20/25/30% during peak hours (17:00-21:00) when customer satisfaction is highest (shown by tip percentage peak at 29.7% at 17:00). Revert to 15/20/25% during off-peak hours. Estimated tip increase: 3-5 percentage points, benefiting drivers.
- Fare Transparency Tools: Develop mobile app showing real-time fare estimates incorporating traffic conditions, time-of-day charges, and typical tip ranges. The 87% distance-fare correlation enables accurate predictions, reducing fare disputes and customer anxiety.
- Group Ride Marketing: Promote flat-rate group pricing for 4-6 passengers leveraging the finding that per-person cost decreases significantly with group size. Target entertainment districts on weekend evenings (19:00-02:00) when group travel is common.
- Payment Modernization: Accelerate credit card adoption (currently 73.8%) through cashless vehicle options and mobile payment integration, improving tip capture and reducing cash handling costs and risks.

Fleet Management Recommendations

- Seasonal Staffing: Expand fleet capacity 10-15% in May and October to capture peak seasonal demand. Contract capacity 8-12% in February and August, offering drivers guaranteed minimum hours to maintain core workforce through troughs.
- Shift Optimization: Weekdays require split-shift model (morning 06:00-10:00, evening 16:00-20:00) or double-shift staffing to cover dual peaks efficiently. Weekend single-shift model (14:00-24:00) optimizes coverage with lower staffing costs.
- Night Operations: Given night hours (23:00-05:00) contribute only 12.1% of revenue despite 11.8% of trips, implement night premium of $2-3 per trip to improve driver earnings and ensure adequate coverage. Target airport runs and nightlife districts.
- Zone-Based Performance Tracking: Monitor driver performance by zone, rewarding drivers who maintain coverage in difficult zones (high-drop-off ratios like Newark) with bonuses while penalizing persistent refusal of short trips or low-fare zones.

Key Performance Indicators

To measure operational success, track the following KPIs monthly:

- Revenue per Vehicle Hour: Target $45-55 during peak hours, $25-35 during offpeak. Current analysis suggests $48 peak, $28 off-peak.
- Average Trip Revenue: Monitor distance-fare correlation (current r=0.87) to detect pricing issues or route inefficiencies.
- Fleet Utilization Rate: Target 75% during weekday peaks, 60% weekends, 35% overnight. Track idle time and reposition fleet accordingly.
- Customer Satisfaction Score: Survey-based metric targeting >4.2/5.0, correlating with tip percentage trends (higher tips indicate satisfaction).
- Driver Retention Rate: Target <15% monthly churn, critical during seasonal troughs to maintain core workforce for peak months.

Conclusion

This comprehensive analysis of 2023 NYC Yellow Taxi data reveals clear patterns and opportunities for optimizing taxi operations in America's largest and most competitive urban transportation market. The data-driven insights span temporal demand patterns, financial performance, customer behaviour, and operational efficiency, providing a robust foundation for strategic decision-making.

The most critical finding is the pronounced 18:00 weekday peak representing the single highest-revenue opportunity, demanding focused fleet positioning and premium pricing during this narrow window. The 12.2% weekday-weekend demand differential and 92% weekday peak-trough variation require flexible staffing models and dynamic pricing to maintain profitability across varied demand conditions. Geographic analysis identifying Midtown Manhattan and Upper East/West Sides as persistent high-demand zones provides clear targeting for fleet deployment.

Financial analysis reveals that while trip volume drives revenue, per-trip value varies by season, time, and distance, creating opportunities for pricing optimization. The strong distance-fare correlation ($r=0.87$) validates current metered pricing but the 13% unexplained variance highlights room for route efficiency improvements and transparent time-based charging. Seasonal patterns showing May and October peaks offset by February and August troughs demand proactive capacity management and cash flow planning.

Customer behaviour insights, particularly the 73.8% credit card adoption and tip percentage peaks during evening hours, enable targeted interventions such as terminal tip suggestion optimization and cashless fleet options. The finding that longer trips generate higher tip percentages creates driver incentives requiring counterbalancing through short-trip base fare adjustments to ensure equitable service across all trip types.

Implementation of these recommendations—dynamic surge pricing during peak hours, zone-based fleet positioning, seasonal capacity adjustments, and customer experience enhancements—should improve revenue per vehicle hour by an estimated 12-18% while enhancing driver retention and customer satisfaction. The data-driven approach ensures resource allocation matches demand patterns, maximizing operational efficiency in a market where competitive advantage derives from superior logistics, pricing precision, and service reliability.

Future analysis should incorporate real-time traffic data for routing optimization, weather impacts on demand patterns, competitive pricing from ride-hailing services, and customer feedback correlation with operational metrics. Continuous monitoring of the recommended KPIs will enable adaptive strategy refinement as market conditions evolve. The foundation established through this 2023 analysis provides the analytical framework and baseline metrics necessary for ongoing operational optimization in NYC's dynamic taxi market.